

Hasan Zaki Rizvi

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EDUCATION

- Curtin University** | Bachelor of Advanced Computer Science Honours | 79 CWA 2024 - Current
- Coursework: Database Systems, Data Structures and Algorithms, Machine Learning, Operating Systems
 - Letter of Commendation (2025)
- Willetton Senior High School** | 96 ATAR 2018 – 2023
- Coursework: English, Methods, Chemistry, Human Biology, Physics

TECHNICAL SKILLS

- Programming Languages:** Java, Python, SQL, C, HTML/CSS
- Tools:** Flask, SQL/PostgreSQL, NumPy, Pandas, RAG, LangChain, FAISS, PyTorch, Git, NeoVim, Linux (Ubuntu, Kali), TensorFlow, Hugging Face, SQLite

PROJECTS

- VALORANT Comms & Pattern Analysis AI Tool** June 2026 – Current
- Python, WhisperX, pyannote, Hugging Face
- Built a local AI-powered VALORANT comms analysis tool that captures microphone, Discord, and in-game voice audio for continuous transcription and post-round review
 - Engineering a speech-processing pipeline using WhisperX, faster-whisper, and pyannote diarization to separate comms by speaker and organise transcripts into structured round-based data
 - Developing machine learning experiments to analyse comms, round context, and player behaviour patterns to investigate whether movement, rotations, and tendencies can be predicted within specific rank brackets
- GenAI for Controlled Human Mobility Data Generation** Feb 2026 – Current
- Python, FAISS, OpenAI API, RAG, NumPy, Pandas, Matplotlib
- Processing large-scale mobility data into structured trajectory segments and generating controlled synthetic afternoon/night trajectories using a RAG pipeline with FAISS retrieval, OpenAI embeddings, and prompt-based generation
- Implementing a RAG-based retrieval system using FAISS, OpenAI embeddings, and cosine similarity/DTW reranking to retrieve similar mobility patterns and guide trajectory generation
 - Developing an evaluation workflow comparing generated trajectories against ground-truth mobility data using DTW, cosine similarity, endpoint distance, grid overlap, and visual trajectory analysis.
- AI Virtual Health Assistant for CALD Healthcare Support** Feb 2025 – Oct 2025
- Python, Flask, LangChain, FAISS, OpenAI API, RAG
- Developed a cross-platform AI healthcare assistant using Flutter, Flask, and SQLite to support culturally and linguistically diverse users with accessible health information
 - Built a LangChain and FAISS-based RAG system for document-grounded healthcare responses, improving answer reliability through prompt engineering, safety filters, and human review workflows
 - Designed structured local data storage and retrieval workflows to manage user queries, healthcare documents, and AI-generated responses across the mobile application

WORK EXPERIENCE

- Seafood Processor** | KB Seafood Oct 2024 – Current
- Processed crayfish products to strict food safety and SOP standards while operating specialised machinery in a high-volume production environment
 - Used WMS for inventory tracking and order fulfilment, maintaining high accuracy and meeting daily production and shipping targets
- Volunteer Store Helper** | Save the Children - Op Shop July 2025 – Current
- Assisted customers in-store, processed transactions and maintained high standards of service
 - Supported store operations including stock receiving, pricing, replenishment, and housekeeping
- Volunteer Helper** | Local Community Events 2019 – 2021
- Operated audio systems for effective sound management & Helped with setup & cleanup

Leadership

- Valorant Premier Team** | Captain & Coach Sept 2023 – Nov 2025
- Developed team strategies to improve competitive performance, including analysing opponents, planning agent compositions, and coordinating utility usage during matches
 - Coached teammates through targeted feedback, VOD reviews, and practice routines while improving communication, teamwork, and in-game decision-making

HOBBIES & LANGUAGES

- Gaming, PC Building, Calisthenics, Gym, Hiking
- English (Native), Urdu (Native), French (Elementary), Arabic (Beginner)